

Problem 5

Problem 5

Problem 1 Given the acceleration function, the initial velocity and initial position of an object moving along a line, find the **position function**.

$$a(t) = 4 \cos t, v(0) = 2, s(0) = 6.$$

Explanation. $v(t) = 4 \sin t + C$.

Letting $t = 0$, we get

$$v(0) = 4 \sin 0 + C \text{ and}$$

$$2 = C.$$

Therefore, $v(t) = 4 \sin t + 2$.

So, $s(t) = -4 \cos t + 2t + C$.

Letting $t = 0$, we get

$$s(0) = -4 \cos 0 + 2(0) + C.$$

So,

$$6 = -4 + C,$$

and $C = 10$.

Therefore,

$$s(t) = -4 \cos t + 2t + 10.$$